Student's Assessment Number……..

THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MASASI DISTRICT COUNCIL
FORM TWO PRE NATIONAL ASSESSMENT

033 **BIOLOGY**

Time: 2:30 Hours **August, 2024**

Instructions

1. This paper consists of sections **A, B** and **C** with a total of **ten (10)** questions.
2. Answer **all** questions. All answers must be written in the spaces provided.
3. Section **A** and **C** carry **fifteen (15)** marks each and section **B** carries **seventy (70)** marks.
4. All writings must be in blue or black ink **except** drawing which must be in pencil.
5. All communication devices, calculators and any unauthorized materials are **not** allowed in the Mock Assessment room.
6. Write your **Pre National Assessment Number** at the top right corner of every page.

FOR ASSESSORS' USE ONLY		
QUESTION NUMBER	**SCORE**	**ASSESSOR'S INITIALS**
1		
2		
3		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

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SECTION A (15 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

(i) Which one of the following branches of Biology deals with the study of insects?

A. Cytology
B. Ecology
C. Entomology
D. Mycology
E. Parasitology

(ii) A pond water drop was observed under a microscope. A unicellular organism was seen moving by using pseudopodia. Which one among the following organisms represents it?

A. Amoeba
B. Euglena
C. Paramecium
D. Plasmodium
E. Spirogyra

(iii) What reason justifies the biology laboratory to be well ventilated?

A. It helps circulation of fumes.
B. It helps the removal of fumes.
C. It provides suffocation to users.
D. It ensures safety to users.
E. Protects users from damages.

(iv) Which one of the following means assists to solve the problem of hiccups?

A. Drinking enough amount of water.
B. Eating enough amount of food.
C. Reducing amount of carbon dioxide.
D. Swallowing large amount of air.
E. Taking small amount of water.

(v) In order to maintain proper health to individuals, the following are important factors to be maintained **except**

A. Have time for break and leisure.
B. Practice regular physical exercises.
C. Reduction of unhealthy food intake.
D. Sleeping in well-ventilated room.
E. Take medical drugs regularly.

(vi) Which one of the following adaptations of palisade mesophyll makes them effective for photosynthesis in plant leaves?

A. Arranged parallel to light source for maximum absorption.
B. Consist of chloroplasts with chlorophylls for capturing light.
C. Direct connected to stomata in order get enough carbon dioxide.
D. Located on lower epidermis to protect from light damage.
E. Irregular shaped cells allowing intercellular space.

(vii) What happens to blood cells when are put into solutions with higher concentration of glucose?

A. Glucose leaves the glucose solution into the blood cells.
B. Glucose enters the blood cells from a glucose solution.
C. Water and glucose leave the blood cells at equal rates.
D. Water enters the blood cells from a glucose solution.
E. Water leaves the blood cells into the glucose solution.

(viii) A cross-sectionally cut stem was immersed in Iodine solution for five minutes and appeared to have a blue black colour. Which part of the stem stained blue black with the given reagent?

A. Cambium
B. Endodermis
C. Pericycle
D. Phloem
E. Xylem

(ix) Respiratory surfaces have features which enable them to be efficient in exchanging respiratory gases. How does the human nasal cavity offer this role?

A. Has a nostril that allows passage of gases.
B. Has cilia for trapping dusts and germs.
C. Has hairs for capturing microorganisms.
D. It is moist to allow secretion of mucus.
E. Produces mucus to warm entering air.

(x) Breathing and respiration occur simultaneously in the body of animals. What properties differentiate the two processes?

(i) Breathing is a mechanical process
(ii) Breathing releases energy
(iii) Respiration is a chemical process
(iv) Respiration uses energy
(v) Respiration uses oxygen

A. (i), (ii) and (iii)
B. (i), (ii) and (iv)
C. (i), (iii) and (iv)
D. (i), (iii) and (v)
E. (i), (iv) and (v)

ANSWERS

QUESTION	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
ANSWERS										

2. Match the adaptations in **List A** with the corresponding parts of the heart in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided

List A	List B
(i) Contracts to pump blood to body parts. (ii) Never gets fatigue from its contraction and relaxation. (iii) Ensures blood flow in one direction towards the body parts. (iv) Separates left and right sides of the heart. (v) Removes wastes from the whole heart.	A Cardiac muscles B Coronary artery C Coronary vein D Semilunar valves E Septum F Sinoatrial node G Ventricular walls H Connection to aorta

ANSWERS

List A	(i)	(ii)	(iii)	(iv)	(v)
List B					

SECTION B (70 Marks)

Answer **all** questions in this section. Each question carries **ten (10)** marks.

3. In the recent years, some secondary school students do not like to study Biology as one of their subjects on their argument that it is difficult to comprehend. Explain **five (5)** importance of studying this subject.

(i) _____

_____

(ii) _____

_____

(iii) _____

_____

(iv) _____

_____

(v) _____

_____

4. The ability of the body to resist against infections and diseases is influenced by some factors.

(a) State **six (6)** factors which affect that body status.

(i) _____

_____

(ii) _____

_____

_____

(iii) _____

_____

_____

(iv)

_____

_____

(v)

_____

_____

(vi)

(b) Differentiate disease from infection.

_____

_____

_____

_____

_____

_____

5. HIV/AIDS incidence in Tanzania is lowering since people are now more aware about its transmission, prevention and treatment of opportunistic diseases.

(a) Identify **two (2)** opportunistic diseases in relation to this infectious disease. (i)

(ii) _____

(b) Explain **four (4)** ways which have been used by Tanzanians to prevent the disease.

(i) _____

_____

(ii) _____

_____

(iii) _____

_____

(iv) _____

_____

6. The living world is composed of prokaryotes and eukaryotes. Provide **four (4)** differences between these kinds of organisms.

(i) _____

_____

(ii) _____

_____

(iii) _____

_____

(iv) _____

_____

7. Lactating mothers and pregnant women should take a balanced diet. Explain **five (5)** reasons why this group of people needs balanced diet. (i)

(ii) _____

(iii) _____

(iv) _____

(v) _____

8. Digestion of food materials in the ileum produces end products by the help of various enzymes. Justify the presence of the enzymes to their roles in this part of the digestive system. Give **five (5)** points.

(i) _____

(ii) _____

(iii) _____

(iv) _____

(v) _____

9. Form Two students at Nata Secondary School carried out a study to determine the interaction of organisms in the school botanical garden and they obtained findings as they appear in **Table 1**. Study carefully the information in **Table 1** then answer the questions that follow.

Organisms	**Numbers**
(i) Locusts	51
(ii) Maize plants	260
(iii) Frogs	19
(iv) Hawks	2

(a) Identify the organisms which are the primary consumers.

_____

_____

_____

_____

_____

(b) Construct a food chain to represent the interaction of the organisms.

(c) State the effect which would occur to this ecosystem when all hawks were killed.

_____

_____

_____

_____

_____

_____

_____

SECTION C (15 Marks)

Answer question **ten (10)**

10. Form Two students in one of the secondary schools in the country agreed that, members of the kingdom Fungi are both useful and unsafe to humans. In **three (3)** points to each case, explain the positions to the students.

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MASASI DISTRICT COUNCIL

FORM TWO PRE NATIONAL ASSESSMENT

033 **BIOLOGY**

MARKING SCHEME

SECTION A (15 Marks)

1.

QUESTION	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
ANSWERS	C	A	D	A	E	B	E	D	C	D

@ 1 mark. (Total 10 marks)

2.

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B	G	A	D	E	C

@ 1 mark. (Total 5 marks)

SECTION B (70 Marks)

3. Importance of studying Biology.
   (i) Helps to understand and care for environment properly as it helps to learn about plants, animals and microorganisms
   (ii) Helps to identify and group living things which makes it easier to learn more about them
   (iii) Helps to appreciate the nature; the interaction between living things and their environment
   (iv) Helps to understand our bodies better
   (v) Helps to improve standards of living and care for animals and plants

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(vi) Helps to answer the fundamental questions; origin life on earth
(vii) Helps scientists to come up with ways to prevent, treat and cure diseases
(viii) Lay foundation for specializations and professions
(ix) It encourages international cooperation through biological studies
(x) Helps to improve food production

Any 5points @ 02=10marks

4. (a) Factors which affect the body immunity.

(i) Poor nutrition
(ii) Lack of vaccination
(iii) Genetic disorder
(iv) Incomplete treatment
(v) Attack by pathogens
(vi) Extreme stress
(vii) Damage to the skin
(viii) Age

Any 6 points @ 01= 06 marks

(b) Difference between disease and infection.

A disease is a condition that interferes with the health of the body while an infection is the disease caused by microorganisms that invade tissues.

Any 1points @ 04 = 04 marks (Total 10marks)

5. (a) Opportunistic diseases in relation to HIV/AIDS.

(i) Tuberculosis
(ii) Pneumonia
(iii) Kaposi's sarcoma
(iv) Skin rashes
(v) Diarrhea

Any 2points @ 02 = 02 marks

(b) Ways which have used by Tanzanians to prevent HIV/AIDS.

(i) Abstinence from sexual intercourse.
(ii) Not sharing clothes, sharp objects with patients/PLWHA.
(iii) Transfusing screened blood which show negative to the virus.
(iv) Wearing protective gears when handling patients/their properties.

Any 4 points @ 02=08 marks Total 10marks

6. Differences between prokaryotes and eukaryotes

Prokaryotes	Eukaryotes
(i) Their cells have no true nuclei	(i) Their cells have nuclei
(ii) They lack membrane bound organelles	(ii) Lack membrane bound organelles
(iii) Their cells have circular DNA material	(iii) Their cells have a linear DNA material
(iv) They are relatively small in size	(iv) They are relatively larger in size
(v) They have few organelles	(v) They have many organelles

Any 5 points @ 02 = 10marks

7. Reasons for Lactating mothers and pregnant women need balanced diet.

(i) Lactating mothers require enough nutrients for themselves and their growing foetus
(ii) Pregnant women require enough nutrients for themselves and their growing babies
(iii) Lactating mothers require food rich in protein to build their breasts and proper functioning of tissues and organs of the babies.
(iv) Pregnant women require food rich in protein to build their uterus and proper functioning of tissues and organs of the foetus.
(v) Pregnant women require food rich in minerals (folic acid) and vitamin B in order to lower risk of birth of babies with hole in the spine (spina bifida).
(vi) Pregnant women require food rich in calcium to develop their foetus' bones.
(vii) Lactating mother require food rich in calcium to develop their baby's bones.
(viii) Pregnant women and lactating mothers require iron for their blood supply needs.
(ix) Pregnant women require food rich in zinc for proper progression of labour during delivery.
(x) Lactating mothers require food rich in zinc for proper growth and development of the baby.
(xi) Pregnant women require food rich in roughages for reduction of constipation.
(xii) Pregnant women and lactating mothers require plenty of water since water is major constituent of breast milk and removes thirsty.

Any 5points @ 02 = 10marks

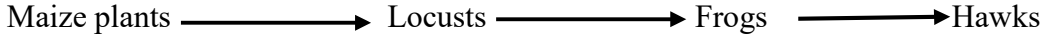
8. Justification of the presence of the enzymes in the ileum.

(i) Lipase breaks down lipids into glycerol and fatty acids.
(ii) Maltase breaks down maltose into glucose.
(iii) Sucrase breaks down sucrose into glucose and fructose.
(iv) Lactase breaks lactose into glucose and galactose.
(v) Peptidase breaks down peptides into amino acids

Any 5points@02=10marks

9. (a) primary consumers are locusts and frogs. **2points@02=04marks**

(b) A diagram of a food chain



02marks

(c) When all hawks are removed from the ecosystem, the number of frogs will increase and the number of locusts will decrease.
2points@02=04marks

Total 10marks

SECTION C (15 Marks)

10. Usefulness and unsafe of members of the kingdom Fungi to humans.

Introduction
Kingdom fungi consists of organisms such as the yeast, mushroom, moulds,
01½marks.

Body
Usefulness
(i) They are used as food
(ii) They are in making bread by raising dough
(iii) They are used in making alcohols by fermenting sugars
(iv) They are used to make commercial enzymes
(v) They are used to make lactic acid, citric acid and cheese
(vi) They are used in production of drugs such antibiotics
(vii) They are used for genetic engineering
(viii) They are used to decompose dead organisms and waste materials
Any 3points@02=06marks

Unsafe

(i) Some cause diseases in animals such candidiasis

(ii) Some cause diseases in plants such bright light

(iii) They cause food spoilage

(iv) Some produce toxins such as aflatoxin

Any 3points@02=06marks

Conclusion

Though fungi shown many uses but have dangers, so they have to carefully handled.

01½marks